

Why the Moon Has Phases (and Locating It)

Phases come from the Moon's position relative to the Sun — not from Earth's shadow.

The real reason for phases

The Sun lights up **one half** of the Moon. As the Moon **revolves around the Earth** (once in ~a month), we see different amounts of that lit half — giving the phases.

- **New Moon:** the Moon is between Earth and Sun, so its lit half faces away from us — we see nothing. It appears **closest to the Sun** in the sky.
- **Full Moon:** the Earth is between Sun and Moon, so the whole lit half faces us. It appears **farthest from the Sun** (rises as the Sun sets).

Phases are NOT caused by Earth's shadow! They come from the changing Sun–Moon–Earth orientation. Earth's shadow on the Moon causes a **lunar eclipse**, which is different.

Eclipses

- **Lunar eclipse** — only on a **full Moon** day (Earth's shadow on Moon).
- **Solar eclipse** — only on a **new Moon** day (Moon between Earth and Sun).

They don't happen every month because the Moon's orbit is slightly tilted.

Locating the Moon

The Moon **rises about 50 minutes later each day**, so it can sometimes be seen in daylight. A **waxing** Moon is easiest to spot at **sunset**, a **waning** Moon at **sunrise**.

Where students lose marks

Trap 1 — Phases are NOT from Earth's shadow. That's a lunar eclipse. Phases are from the Sun–Moon–Earth positions.

Trap 2 — Lunar eclipse = full Moon; solar eclipse = new Moon. Don't swap.

Trap 3 — The Moon rises ~50 minutes later each day, so it doesn't always rise at sunset — that's why it can be seen by day.

Model answers — write them like this

Example A. Why can the Moon sometimes be seen during the day?

Step 1: The Moon rises about 50 minutes later each day.

Step 2: On many days it rises in the afternoon, so it is up while the Sun is still out.

∴ Its shifting rise time means it is often above the horizon in daylight.

Example B. A friend says the Moon's phases are caused by Earth's shadow. Correct this.

Step 1: Phases come from the changing Sun–Moon–Earth orientation as the Moon orbits.

Step 2: Earth's shadow on the Moon causes a lunar eclipse, not phases.

∴ Phases are due to position, not Earth's shadow.

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